

Paper Manager pack — MolmoAct2 (Oct 28)

Paper: MolmoAct2: Action Reasoning Models for Real-World Deployment

arXiv: <https://arxiv.org/abs/2605.02881>

Code: allenai/molmoact2 (Ai2)

PDF: /workspace/papers/molmoact2-2605.02881.pdf

Session: Oct 28 · due 9:30 AM · **all seats empty** — full five packs

Sources: Summarizer + Senior Research (/workspace/molmoact2-quiet/DELIVERABLE.md)

1. One-liner

Open **Action Reasoning Model (ARM)**: geometric **System-2** (adaptive depth tokens in Think) before a **per-layer-KV DiT FM** action expert — deployable open alternative to closed π -series.

2. CRITICAL System-2 distinctions (do NOT conflate)

Outer loop	What it is	What it is NOT
MolmoAct2 Think	Adaptive depth / geometry codes (10×10)	$\pi_{0.5}$ HL semantic subtask strings
$\pi_{0.5}$ HL	Semantic subtask language	Depth lattice
BEHAVIOR'25 System-2	Task-progress stage IDs + voting	Scene geometry

All three call an FM chunk expert (System 1). **TENTATIVE:** BEHAVIOR may want **depth + stage** complementary.

FALSE claims: “has $\pi_{0.5}$ HL” / “has RTC” / “has BEHAVIOR S2 voting.”

3. Stack snapshot

- Backbone: Molmo2-ER (4B LLM + SigLIP2)

- Curriculum: FAST discrete AR pretrain → FM expert post-train (KI) → FT
 - FM expert ~621M, L=36, per-layer KV; infer N=10 Euler
 - Think: 100 depth codes; regen if RGB cosine <0.996
 - Rates: ~55.8 Hz Act2 / ~12.7 Hz Think @ H100 H=10 + CUDA Graphs
 - Claims: LIBERO **97.2 / Think 98.1** ($\pi_{0.5}$ 96.9); MolmoSpaces **37.7** vs **34.5**; DROID OOD **87.1%**
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4. Motion / control hierarchy

Layer	What	Interface
L0	SigLIP2+LLM multi-cam	tokens + per-layer K/V
L1 System-2 (opt Think)	AR **depth codes**	depth → VLM context → expert KV
L2 Motor	DiT FM; H≈Hz; N=10 Euler	continuous joints

Closed-loop: chunked FM (open-loop within chunk, replan) — same family as $\pi_0/\pi_{0.5}/DP$. Paper often open-loop; Comet/BEHAVIOR prefer **receding horizon** — don’t cargo-cult open-loop on long-horizon household tasks. **RTC absent** in PDF — latency via FM cache + CUDA Graphs + Think reuse.

5. Role packs

Stakeholder (18+3)

Open ARM with weights/code/data; competitive with $\pi_{0.5}$ on paper protocols; openness vs π opacity. Risks: Spatial OOD; OOTB mainly SO-100/Franka. Use as open FM-VLA+ARM baseline when π gated.

Scientific Reviewer (8+3)

Credible systems paper; deltas = per-layer KV + adaptive depth. Caveats: Think threshold unablated; $\pi_{0.5}/GR00T$ not always compute-matched; “>GPT-5” is ER-VLM not robot SR. Verdict: open ARM + FM interface advances; not a theorem paper.

Empiricist (11+3)

Priority	Experiment
P0	Think vs flat continuous; vs $\pi_{0.5}$ on LIBERO/DROID proxy; log SR + ms/chunk + Hz
P1	Per-layer KV; K flow samples; adaptive depth; horizon×open-loop vs receding

Defer	Full pre/post; real YAM 720h
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Tattoo: LIBERO 97.2/98.1; Spaces 37.7; DROID 87.1; KV 95.9>hidden 94.0; 55.79/12.71 Hz. Env: /workspace/molmoact2-quiet/env_outline/.

Archaeologist (11+3)

Molmo/Molmo2 → Molmo2-ER → MolmoAct → **MolmoAct2** (FAST+per-layer KV FM) → Think. Parallel PI: $\pi_0 \rightarrow \text{FAST} \rightarrow \pi_{0.5} \rightarrow \text{KI} \rightarrow \text{RTC} \rightarrow \text{BEHAVIOR S2}$. Hybrid FAST → FM curriculum like $\pi_{0.5}$. New: ER; YAM data; per-layer KV; adaptive depth.

Visionary (11+3) — 2026 BEHAVIOR

- 1. Keep FM low-level (discrete AR slower — aligns FAST≠FM-at-deploy)
- 2. Choose S2 deliberately: semantic HL/stages vs depth ARM vs **hybrid**
- 3. Open substrate when π gated; don't assume OOTB wins BEHAVIOR without stage tracking + RTC
- 4. **RTC wrap** on Molmo FM = high-leverage follow-on (**TENTATIVE**)

Skip Discord/Drive.

6. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Open ARM vs π opacity
Reviewer	8+3	Systems deltas + cite hygiene
Empiricist	11+3	Think on/off + Hz/latency
Archaeologist	11+3	Parallel Molmo vs PI lineages
Visionary	11+3	Three outer loops for 2026

Spine: geometric S2 ≠ semantic HL ≠ stage voting; FM motor. PDF for Mac.

GitHub title: [ECE 605] - MolmoAct2 Making Club Pack

Paths

Artifact	Path
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This pack	`/workspace/molmoact2-quiet/KANTA_PACK.md`
Senior Research	`/workspace/molmoact2-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/molmoact2-2605.02881-summary.md`
Paper PDF	`/workspace/papers/molmoact2-2605.02881.pdf`